

A comprehensive introduction to Text Analytics in HR

Problem we're solving

How do you know what your employees think?

- Better information = Better decisions
- Employees have best ideas because they're closest to the problem
- Communication via Chinese Whispers is a significant barrier
- Surveys are used because they're cheap, not because they're effective
- Asking open questions has always been expensive to scale.

Benefits of open text feedback

Open questions ensure that the best quality of data is being collected, analysed and acted upon.

For the employee

**“I can say
exactly what I
think”**

Open questions provide a better experience letting employees share their thoughts in their own words.

It's quick

Most Workometry feedback requests take less than 5 minutes to complete.

**It's mobile-
friendly**

Employees use whatever device they want. Over 50% use a smartphone.

**Change
comes quickly**

The richer, contextual feedback means firms don't need additional detail. Therefore they can act more quickly.

For the firm

**Ask about any
topic**

Don't worry about validating complex survey designs: just ask a few great open questions.

**The data is
rich & has
context**

The answers you'll get will show you why it's important and how to act effectively.

**Quickly
identify key
themes**

Workometry's sophisticated analytics highlight key themes and let you dive into details.

**Segment &
target**

Thoughts aren't distributed uniformly: Workometry links to your data to identify target groups.

Ability to make changes quickly

Text, analysed correctly can identify be a real driver for meaningful change.

From...

in the next survey

...To

“Change the music.
It’s so moody and
depressing and totally
not our customer.
We have had several
complaints from
customers and the
team”

“Since the last survey,
they have changed the
music and it creates a
better work place for
staff and customers”

Examples of questions our clients have asked

“How can we improve customer experience in our stores?”

“What could we do to improve the performance review process?”

“What could we do to improve your working life here?”

“What ideas do you have to improve the Christmas sales period?”

“How could HR improve the services they provide you as a manager?”

“If you were our Brand Director, what would you do first?”

What is good enough?

The more you are going to rely on text, the better the coding has to be. You should be able to understand the comment just by the label.



A word cloud of negative terms for code comments, arranged in a circular pattern. The words are: Useless (blue), Superficial (green), Pretty (orange), Pointless (green), Chartjunk (light green), trivial (brown), Rubbish (dark red), and Overused (yellow). The word 'Wordclouds' is also present in blue, positioned diagonally across the center.

Useless

Superficial

Pretty

Pointless

Chartjunk

trivial

Rubbish

Overused

Wordclouds

Finding meaningful topics

We aim to highlight meaningful topics that increase understanding. We find these from your answers, not a standard topic library. More answers mean more insight.



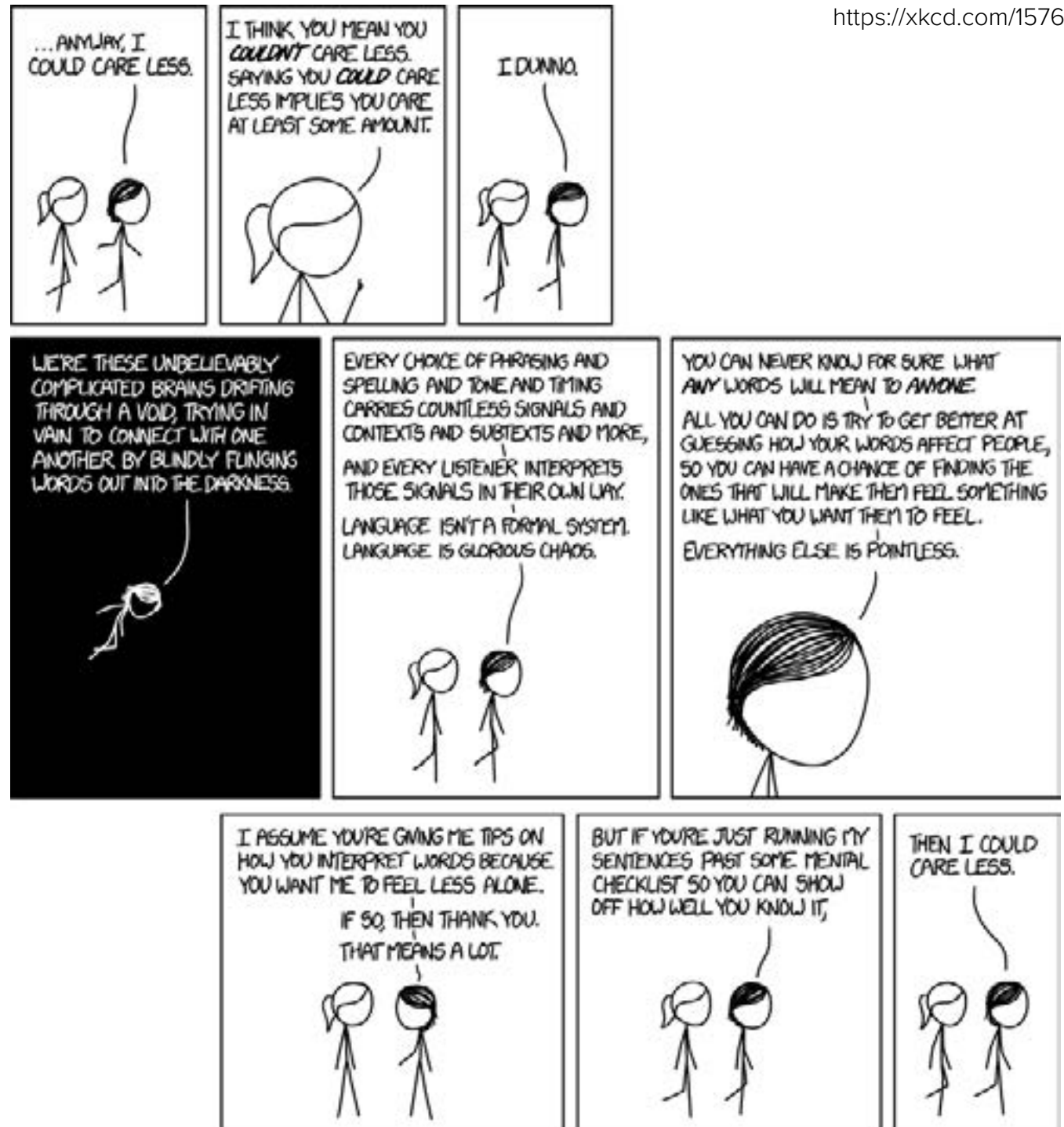
OVERVIEW OF TEXT ANALYTICS

Use cases, history, challenges

Understanding text

Hint: It's difficult

<https://xkcd.com/1576/>



Most data is unstructured

Most of the talk about Big Data was about the rise in unstructured data

According to Forrester (June 2018) enterprises have huge volumes of
unstructured data

61% > 100 terabytes,

12% > 5 petabytes

History of text analytics

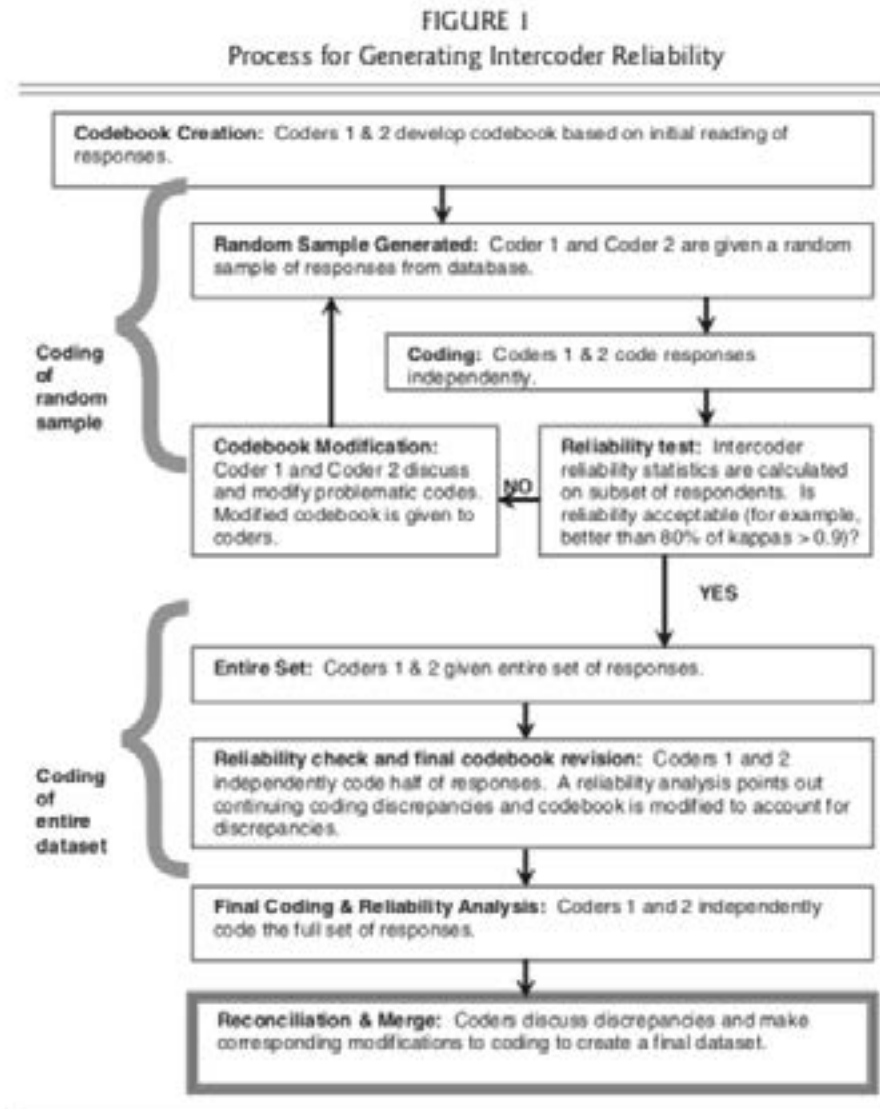
3 phases of text analytics. Data / Computing availability has been a key driver for promoting the adoption of new approaches

**NLP / Programmed
Grammatical
Approaches**

Statistical Approaches

**Deep Learning / Word
Embeddings**

The traditional 'manual' text coding process



Text analysis historically expensive

HR has been trying to avoid capturing and analysing text.

POSITIVES

- Contains richness & Granularity
- Enables people to add what they need / want to say
- Easier to design a good UI

NEGATIVES

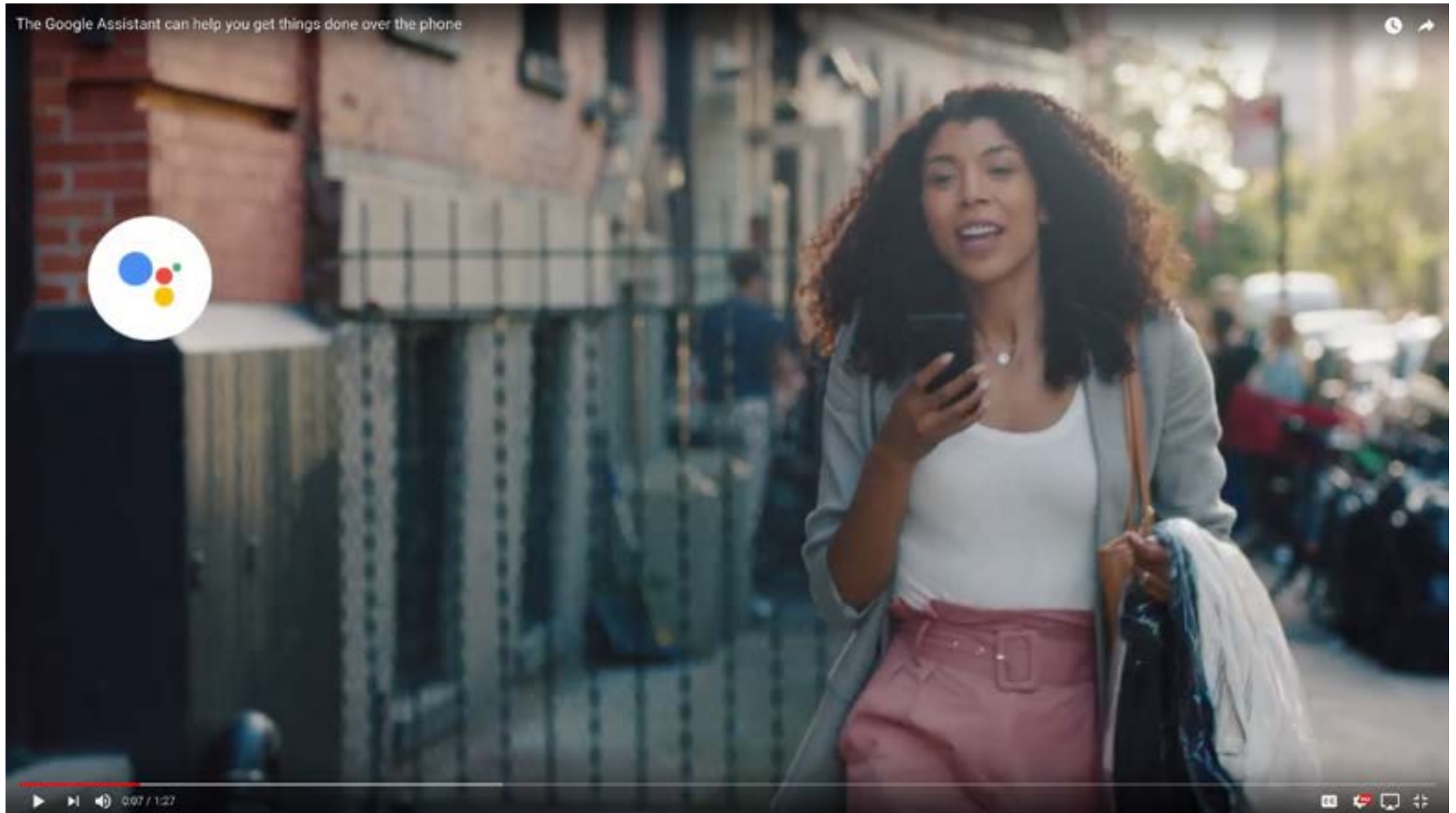
- Messy - typos, multiple-wordings
- Introduction of subjectivity - potential bias
- Risk of litigation
- Difficult to report

Use cases of text analytics

Text analytics are everywhere, and could be the future of computing

- Text Classification
- Language Modeling
- Speech Recognition
- Caption Generation
- Machine Translation
- Document Summarization
- Question Answering

Assistants - Powered by text analysis



Translations

 Microsoft 365 Azure Office 365 Dynamics 365 SQL Windows 10 More ▾

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Achieving Human Parity on Automatic Chinese to English News Translation

March 12, 2018

[Download PDF](#)[View Link](#)[BibTex](#)

Authors

[Hany Hassan Awadalla](#)[Anthony Aue](#)[Chang Chen](#)[Vishal Chaudhary](#)[Abstract](#) [Related Info](#)

Machine translation has made rapid advances in recent years. Millions of people are using it today in online translation systems and mobile applications in order to communicate across language barriers. The question naturally arises whether such systems can approach or achieve parity with human translations. In this paper, we first address the problem of how to define and accurately measure human parity in translation. We then describe Microsoft's machine translation system and measure the quality of its translations on the widely used WMT 2017 news translation task from Chinese to English. We find that our latest neural machine translation system has reached a new state-of-the-art, and that the translation quality is at human parity when compared to professional human translations. We also find that it significantly exceeds the quality of crowd-sourced non-professional translations.

Image to text

Automatically generated text, using a Corpus of Romantic Novels.



Generated story about image

Model: Romantic Novels

"We had to act much of the group, despite the fact that people were surrounded by thousands of people . A few more times, they were beautiful, and I wondered what was going on in New York.

I didn't want to tear my eyes away from Bill. I had no idea what he meant to me. The rest of the crowd gathered around us, and for a moment. I was pretty sure she was the only person in the entire world. He sounded so relieved and heavy , as if it took hundreds of people standing upon them."

HR use cases of text analytics

There are lots of use cases, most of which fit into these groups

Understanding Job
Descriptions / CVs

Understanding
employees' opinions,
needs and ideas

Automating HR
Operations

CV / Job Matching

TextKernel (and other similar products) can match on Job Description / CV pair without recruiters needing to build complex searches.

The screenshot displays the 'Match' interface by Textkernel, a job search platform. The interface is divided into a left sidebar for filters and a main content area for search results.

Left Sidebar Filters:

- Tags** (dropdown)
- Positions** (dropdown)
- Profession Classification** (dropdown)
- City** (dropdown)
 - Postal/ZIP code or city:
 - Radius: +50 km
 - Selection: Nice to have (radio) / Must have (radio)
- Education Classification** (dropdown)
- Workgevers** (dropdown)
- Years of Experience** (dropdown)
- Skills** (dropdown)
- Last Modified** (dropdown)
 - ☐ today (0)
 - ☐ last three days (1)
 - ☐ last week (1)
 - ☐ last 2 weeks (3)
 - ☐ last month (4)
 - ☐ last 3 months (9)
 - ☐ last 6 months (12)

Main Content Area:

- Search Bar:** Includes a search icon and a 'Search!' button.
- Filters:**
 - Recent Positions:** [linux system engineer] x
 - Profession:** Linux Engineer x
 - Job Category:** Information and Communication Technology x
 - Job Subcategory:** System Developers and Analysts x
 - Education:** Bachelor x
 - IT Skills:** configuration management x, Puppet +4 x, SaltStack +7 x, TCP/IP networks x, Tomcat +2 x, PostgreSQL +8 x, MySQL +8 x, Cassandra +10 x, Artificial Intelligence x, NLP x, DevOps x, Linux +20 x, network infrastructure x, SaaS x, infrastructure x, automation x, Computer Science x, network +10 x, programming +5 x, TCP/IP +5 x, Nagios +15 x, Jenkins +33 x, continuous deployment x, high-availability x, database +24 x, distributed storage x, Ceph +8 x
 - Years of Experience:** 3 to 5 years x / 6 to 10 years x
 - Full text:** administration +20 x, reputation x, architecture x, degree of customization x, security needs x, professional services x, engineering +6 x, troubleshooting of TCP x, IP networks x, Java web applications x, monitoring x, deployment tools x
- Results Summary:** Mijne kandidaten (46K) with social media icons (LinkedIn, Twitter, Facebook, etc.).
- Actions:** Add to Shortlist, Compare Profiles.
- Relevance:** 45,609 results.
- Results List:**
 - Linux System Engineer
 - Tanisha Halin / DTV System Engineer / Liberty Global
 - Vol Brander / Linux Engineer / adBev Classified Group

Bots

Automating common processes to reduce cost

Saberr
Work well together.

Welcome!

Today we're going to look at setting some 'norms' for the team. We'll learn what 'norms' are all about in a moment.

But first, let's warm up a bit.



If you could pick up a new skill in an instant what would it be? Spend five minutes going around the team. Doctor Demo, you start. Hit Continue once everyone is done.



Team champ, read this out loud 

Context is vital for understanding text information

Generic text algorithms fail to provide contextual meaning to comments. We look at text in 3 ways:

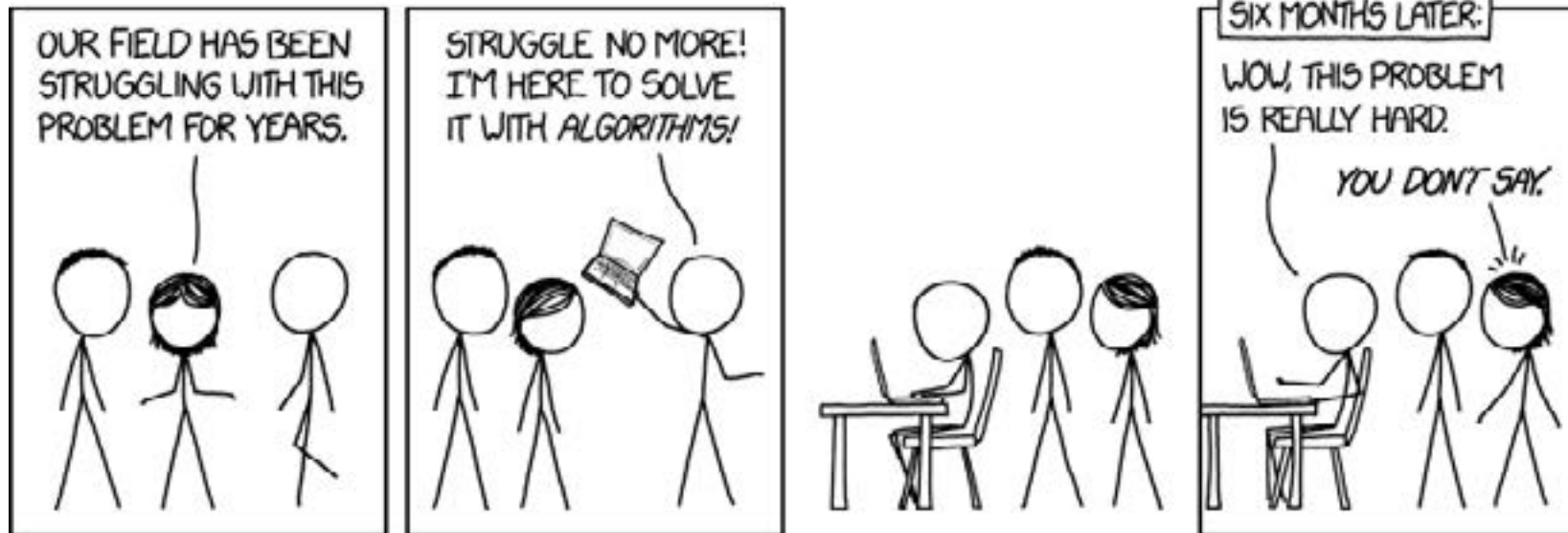
The context of the Organization

The context of the Question

The context of the sentence structure

There are no easy solutions

But we're getting good at solving specific problems



Voice Assistants

Apple's approach to let Siri do everything is much more difficult to achieve than Amazon's approach to opening Alexa up to developers to build specific models.



Very good, at narrow use-cases

The Google Assistant can help you get things done over the phone



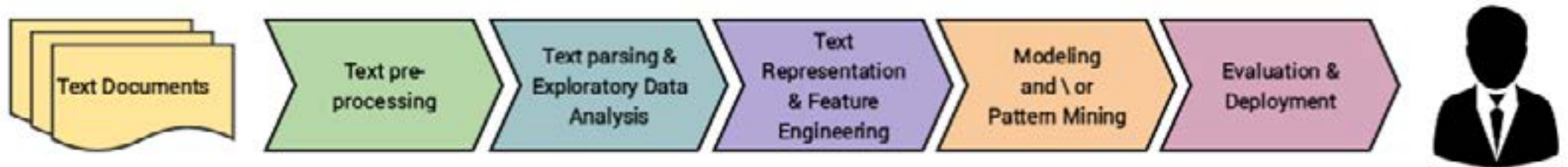
The system only works when it has been trained in extremely narrow contexts, or “domains”, ... since the range of the language used is limited and it is easier to predict what people will say next.

▶ ⏮ 🔊 0:07 / 1:27

⏸ ⏮ ⏭ 🔊

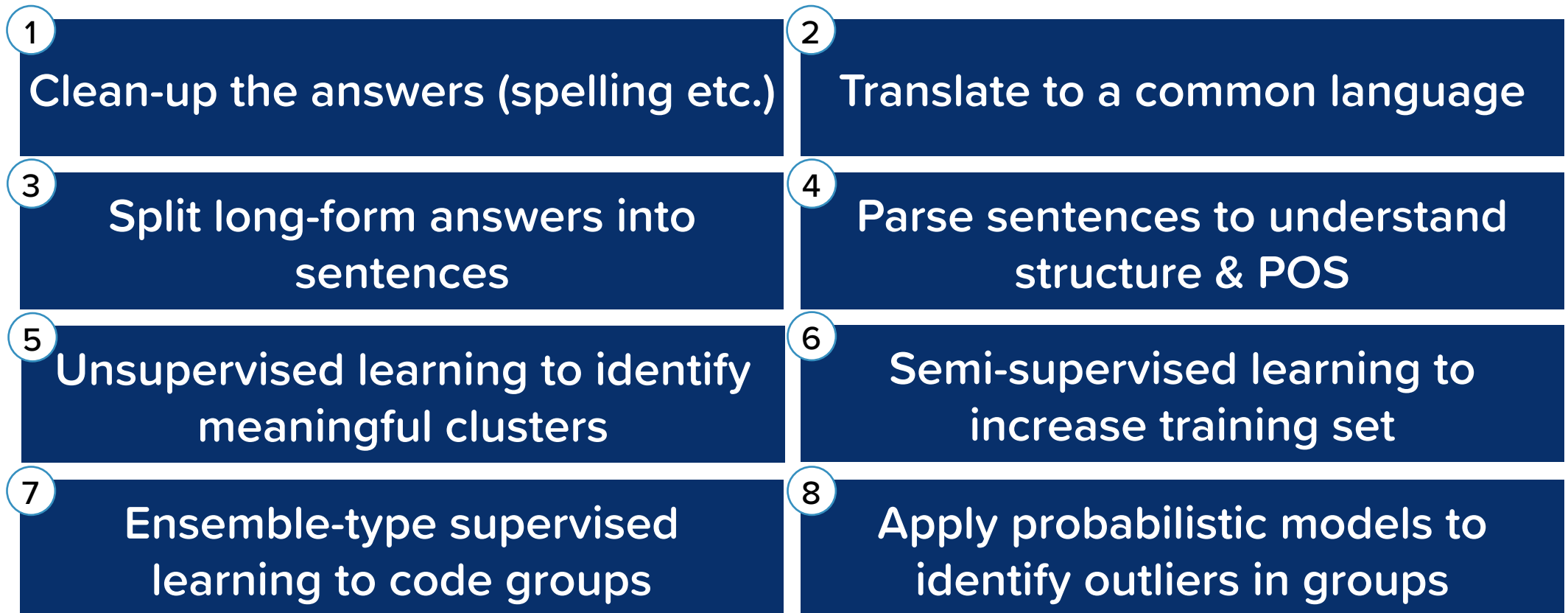
A process for understanding text

We can apply a 'standard' data science methodology to understand most texts



How we code text

A high-level summary of our approach



Throughout, algorithms revert to human to clarify uncertainty.

Parsing trees to understand and segment sentences

Parse trees help you understand the structure of text in a graph (tree) format.

Try the API

Text Analytics is a key part of People Analytics

ANALYZE

[See supported languages](#)

Entities

Sentiment

Syntax

Categories

☒ Dependency

☒ Parse Label

☒ Part of Speech

☒ Lemma

☒ Morphology

nn

Text

NOUN

number=SINGULAR
proper=PROPER

nsubj

Analytics

NOUN

number=SINGULAR
proper=PROPER

root

is
be

VERB

mood=INDICATIVE
number=SINGULAR
person=THIRD
tense=PRESENT

det

a

DET

amod

key

ADJ

attr

part

NOUN

number=SINGULAR

prep

of

ADP

nn

People

NOUN

number=PLURAL
proper=PROPER

pobj

Analytics

NOUN

number=SINGULAR
proper=PROPER

Named Entity Recognition

Try the API

The head of UBS' digital strategy in the U.S. is moving to a rival. The departure comes as an upset for the bank's robo-banking efforts.

[See supported languages](#)

Entities

Sentiment

Syntax

Categories

The (head)₁ of (UBS)₂ digital (strategy)₃ in the (U.S.)₅ is moving to a (rival)₄. The (departure)₆ comes as an (upset)₉ for the (bank)₇'s (robo-banking efforts)₈.

1. head Sentiment: Score 0 Magnitude 0 Sallience: 0.34 PERSON	2. UBS Sentiment: Score 0 Magnitude 0 Wikipedia Article Sallience: 0.22 ORGANIZATION
3. strategy Sentiment: Score 0 Magnitude 0 Sallience: 0.14 OTHER	4. rival Sentiment: Score 0 Magnitude 0 Sallience: 0.10 ORGANIZATION
5. U.S. Sentiment: Score 0 Magnitude 0 Wikipedia Article Sallience: 0.06 LOCATION	6. departure Sentiment: Score -0.6 Magnitude 0.6 Sallience: 0.06 EVENT
7. bank Sentiment: Score 0 Magnitude 0 Sallience: 0.04 ORGANIZATION	8. robo-banking efforts Sentiment: Score -0.1 Magnitude 0.1 Sallience: 0.02 OTHER
9. upset Sentiment: Score -0.6 Magnitude 0.6 Sallience: 0.02 OTHER	

Stemming & Lemmatization

Simplifying the text to core components is useful in many applications

“My system keeps crashing! his crashed yesterday, ours crashes daily”

Stemming

‘My system keep crash hi crash
yesterday, our crash daili’

Lemmatization

‘My system keep crash ! his crash
yesterday , ours crash daily’

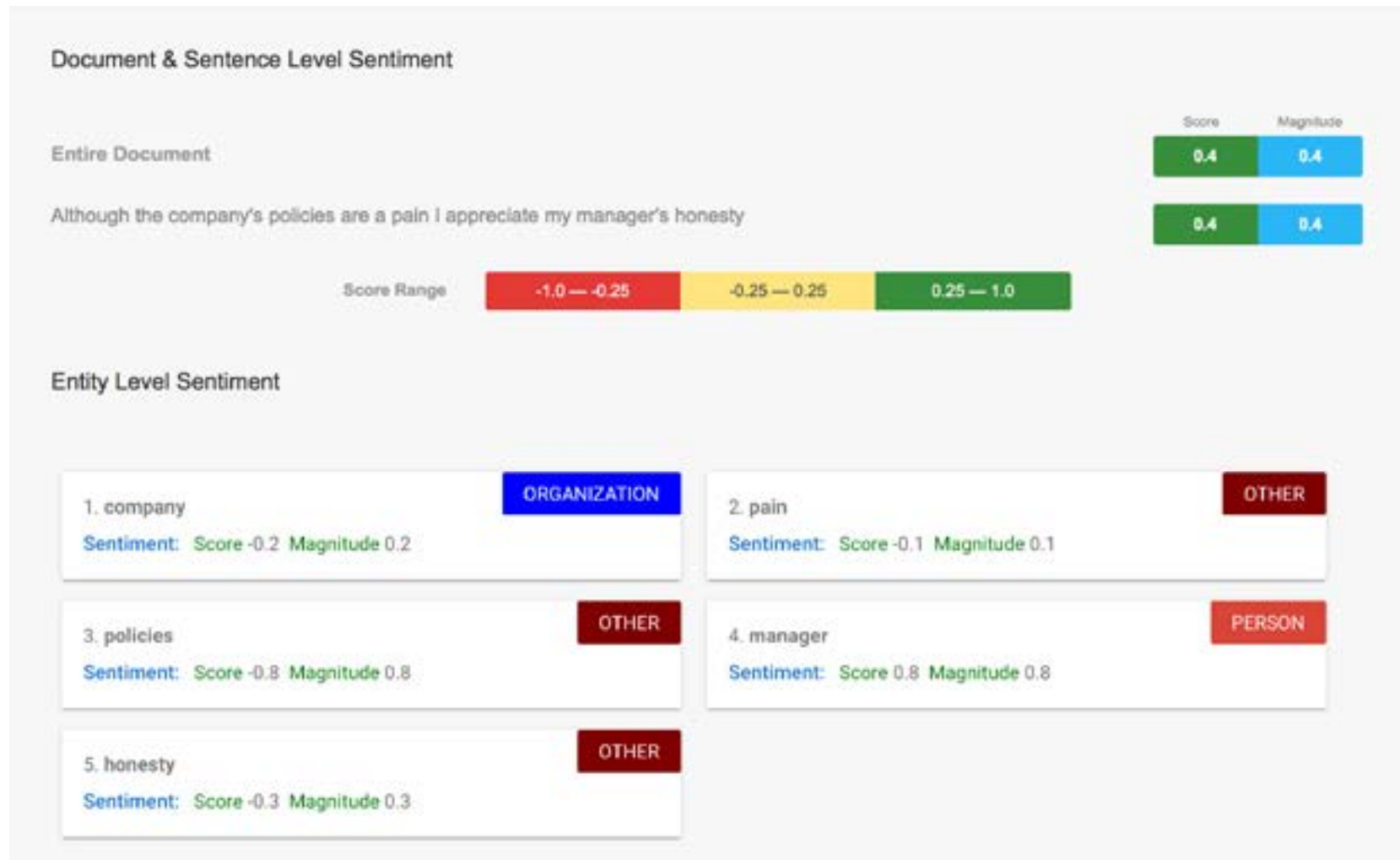
Parts of Speech tagging (POS)

Understand what each part of a sentence means



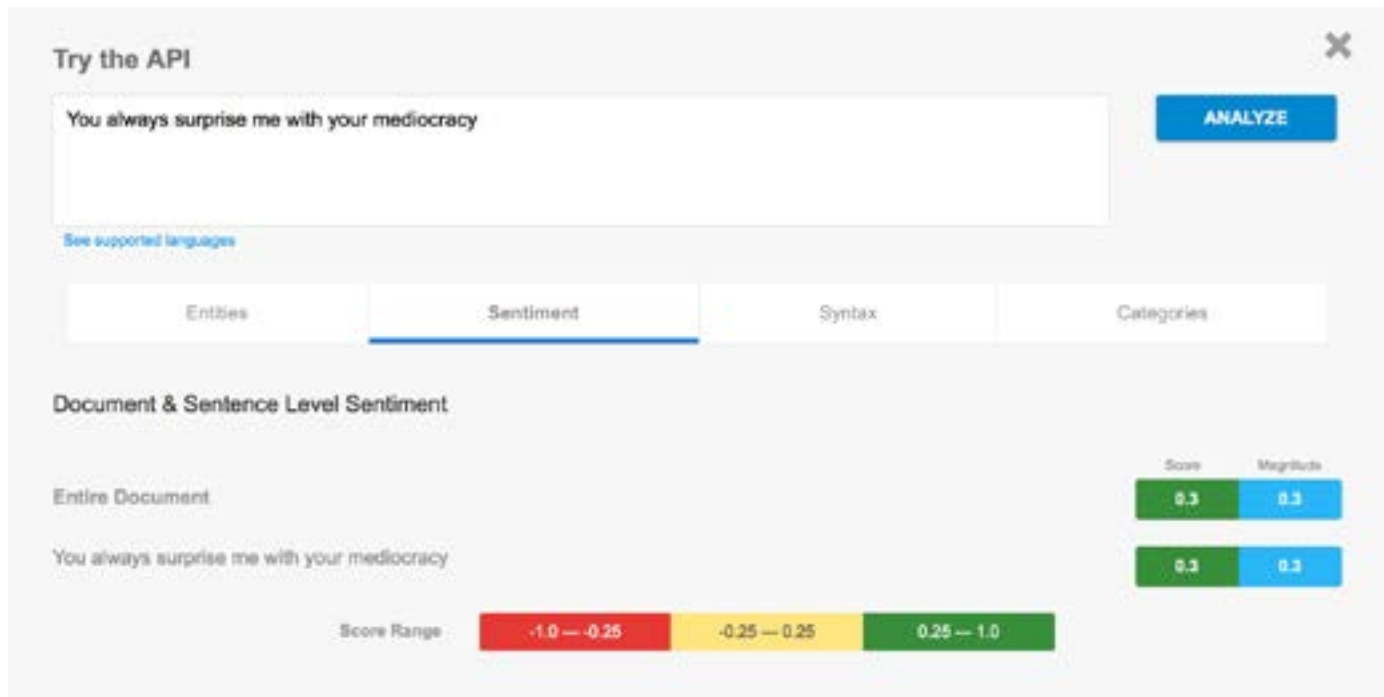
Sentiment Analysis

A useful fall-back solution



Sentiment Analysis cont.

Current state-of-the-art is about 65% accuracy



Tone analysis

Scoring is possible on a range of emotions, not just on sentiment.

Document-level

Tones

Angry ☐ Fear ☐ Joy ☐ Sadness ☒ Analytical ☒ Confident ☐ Tentative ☐

[View JSON](#)

Sentence-level

Identify sentences with stronger tones in context or sorted by score. Highlighted sentences indicate the likelihood of a tone present. If more than one tone is present, the stronger one is shown. Click on a sentence to see a breakdown of all tones.

Tones

Analytical

Confident

Sadness

In context

Sadness: Indicates a feeling of loss and disadvantage. When a person can be observed to be quiet, less energetic and withdrawn, it may be inferred that sadness exists.

< .5 **.5 - .75** > .75

None Strong

Also it not tore on customers.

In addition the quality of store closing has gone down hill and the customers are returning a lot items because of that.

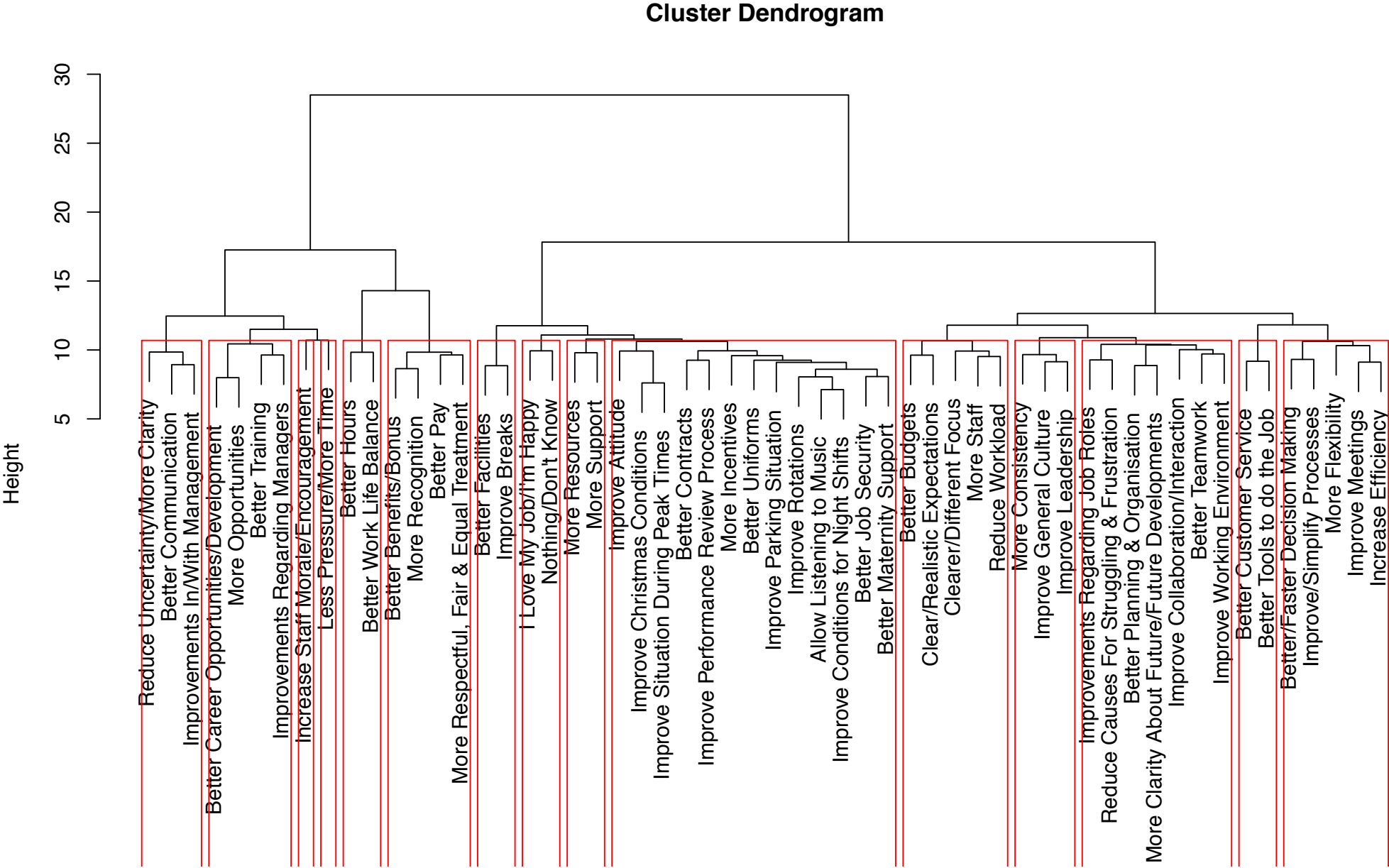
Also working for 8 hours without a break is really had for a lot of staff.

I love working for this firm but at times it's really stressful when communications are bad between staff and head office.

If possible as we'll in delivery can they please send more 3 and 6s, I say this every week and nothing changes.

Customers are compling that we never have those sizes in store Because we get 1

Clustering



Text Classification

TEXT EXPLORER

Please select a topic to see the comments:

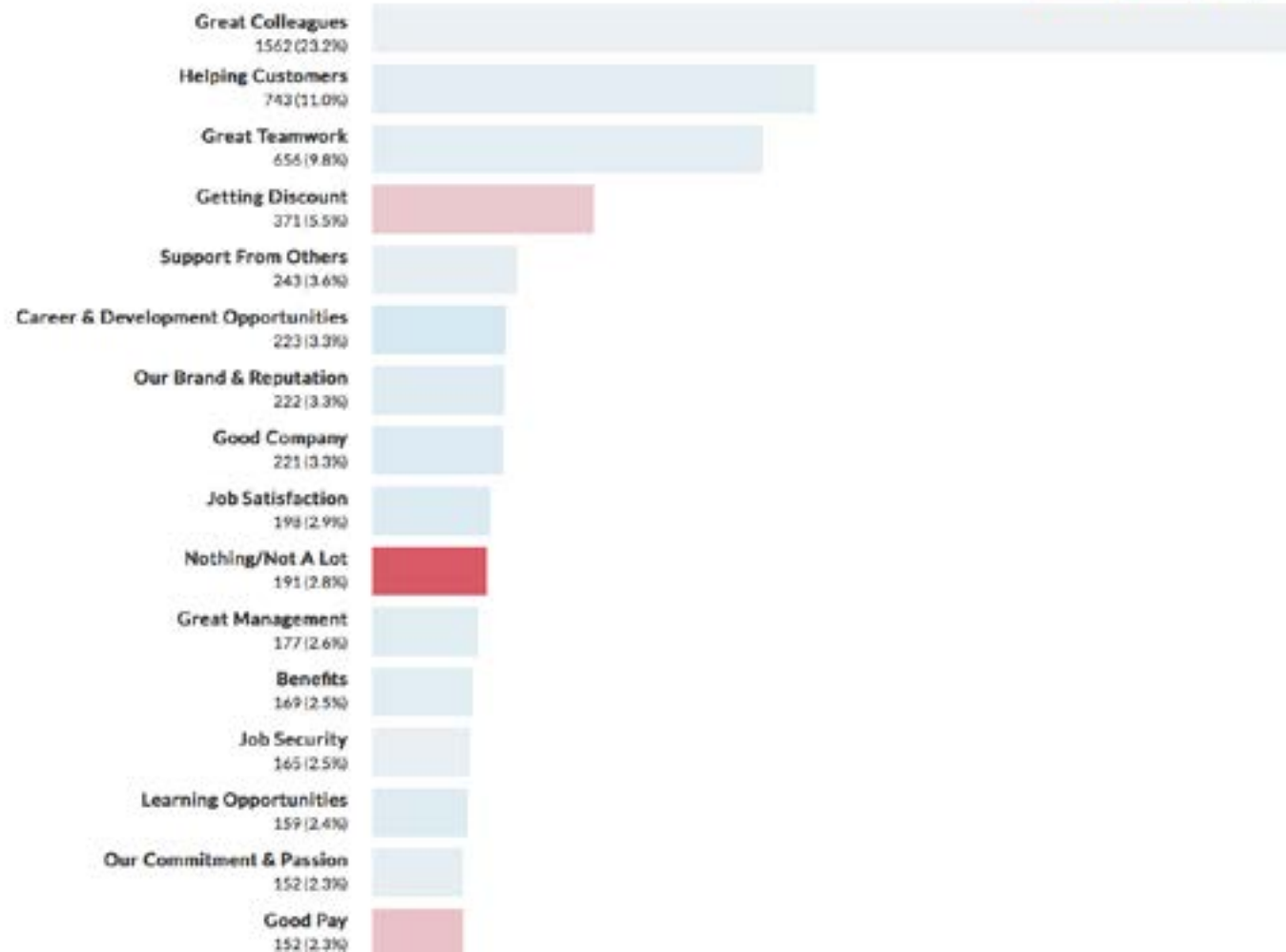
What do you think is the best thing about working at BigRetailer?

LEGEND

Below average Average Above average

Save data

Save graph



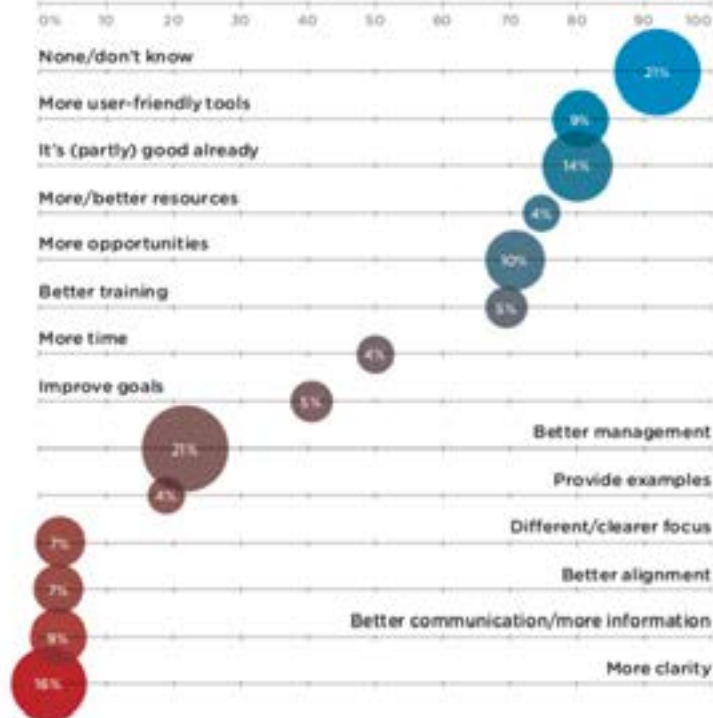
Text Classification

Emerging themes

In response to the question:
What improvements or other tools/resources are needed to support effective goal-setting?

● Proportion of staff who commented on this theme

Probability that the staff member who commented on this theme had a career development conversation

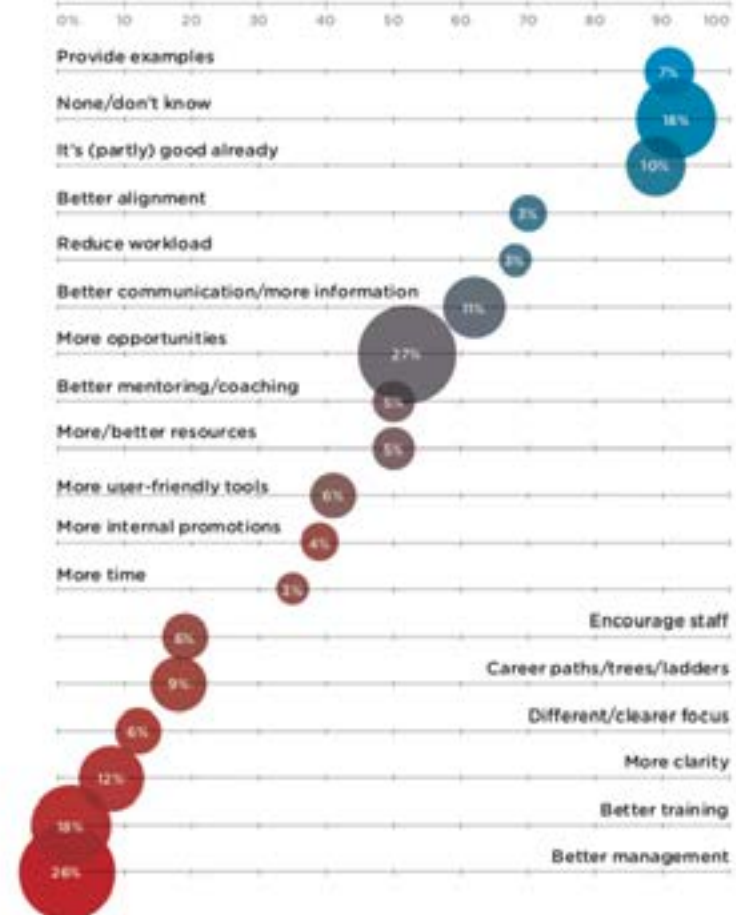


Emerging themes

In response to the question:
What improvements or other tools/resources are needed to support effective career development?

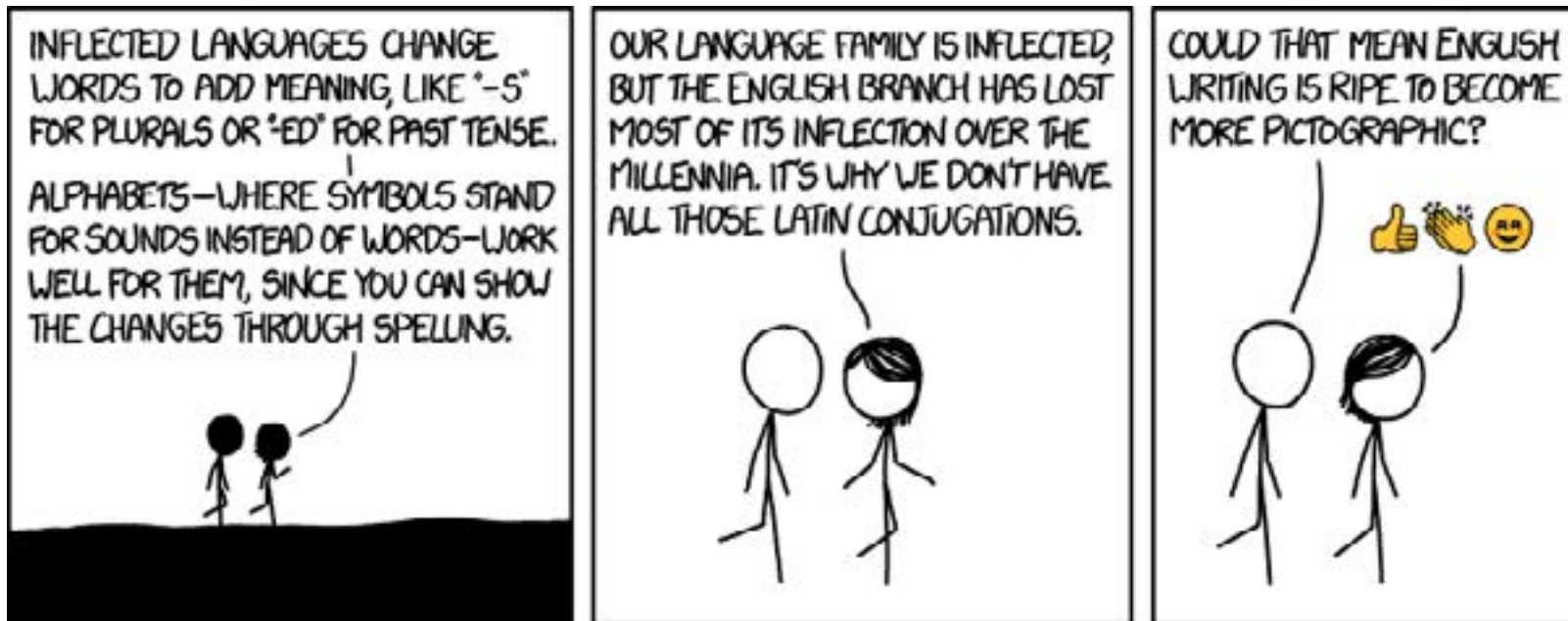
● Proportion of staff who commented on this theme

Probability that the staff member who commented on this theme had a career development conversation



More than just understanding the words

We need to find a way to deal with the way people communicate, especially in short texts.



Need to understand the semantic meaning

“If you could do one thing to improve working for COMPANY what would it be?”

“more money, less hours!”

“Not being salary capped after
30 + years of loyal service”

“Competitive salary increases
for internal movement, low pay
raises encourage outward moves
rather than talent retention”

“Better pay”

“Better wages”

“if I could I would give a pay rise
to the living wage amount and
not just the minimum wage, we
do work hard for our company.”

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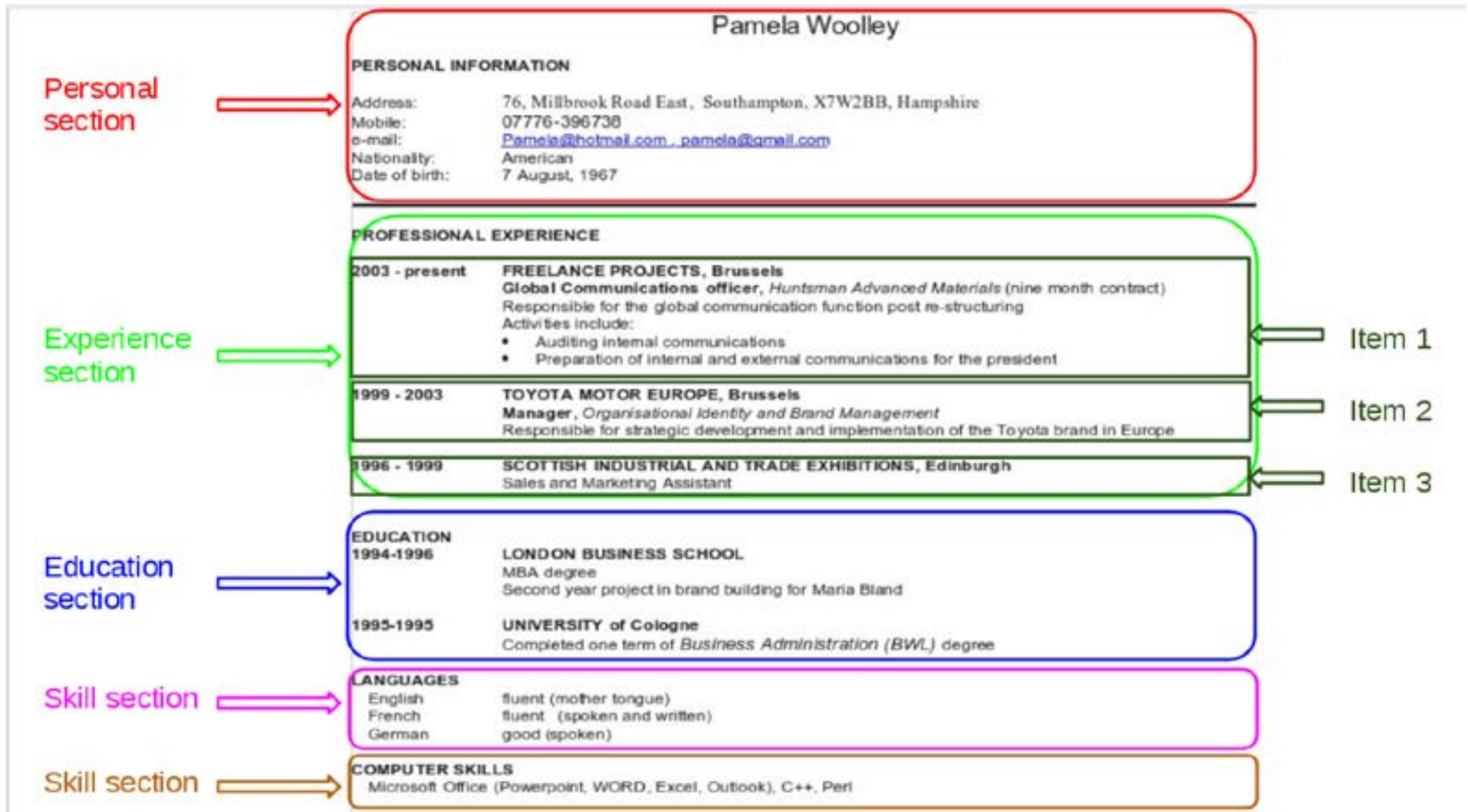
“Better pay”

“Ha Ha. A raise”

“Better wages”

“if I could I would give a pay rise
to the living wage amount and
not just the minimum wage, we
do work hard for our company.”

CV / Job Parsing - Importance of 'Zoning'



Increasing your training set

You always want better training sets



Increasing your training set

You always want better training sets



DEEP LEARNING & WORD EMBEDDINGS

Magic?

History of text analytics

3 phases of text analytics. Data / Computing availability has been a key driver for promoting the adoption of new approaches

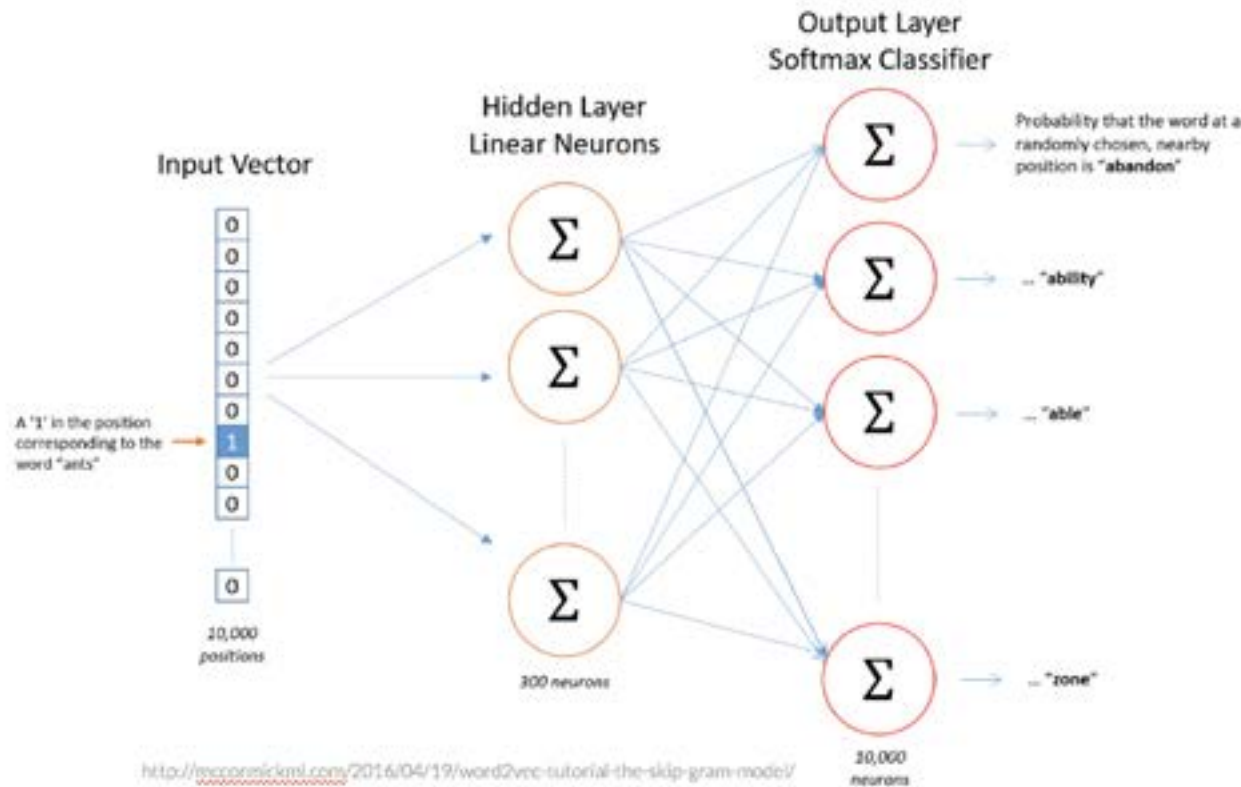
**NLP / Programmed
Grammatical
Approaches**

Statistical Approaches

**Deep Learning / Word
Embeddings**

Word Embeddings - driving current growth

Word Embeddings use deep learning to map words to n-dimensional vectors (where n is



“Skip-Gram”
With one-hot encoded centre word, we can predict context words.

Hidden layer creates embeddings

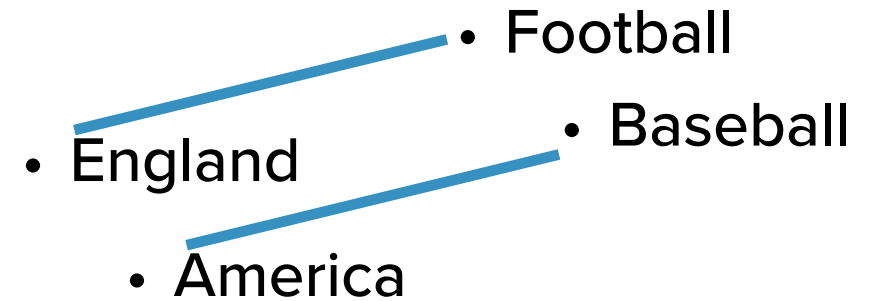
Word Embeddings

Dimensionality reduction can give you an idea of what is happening



Text embeddings learns relationships

Vectors can be subtracted to show learned relationships.

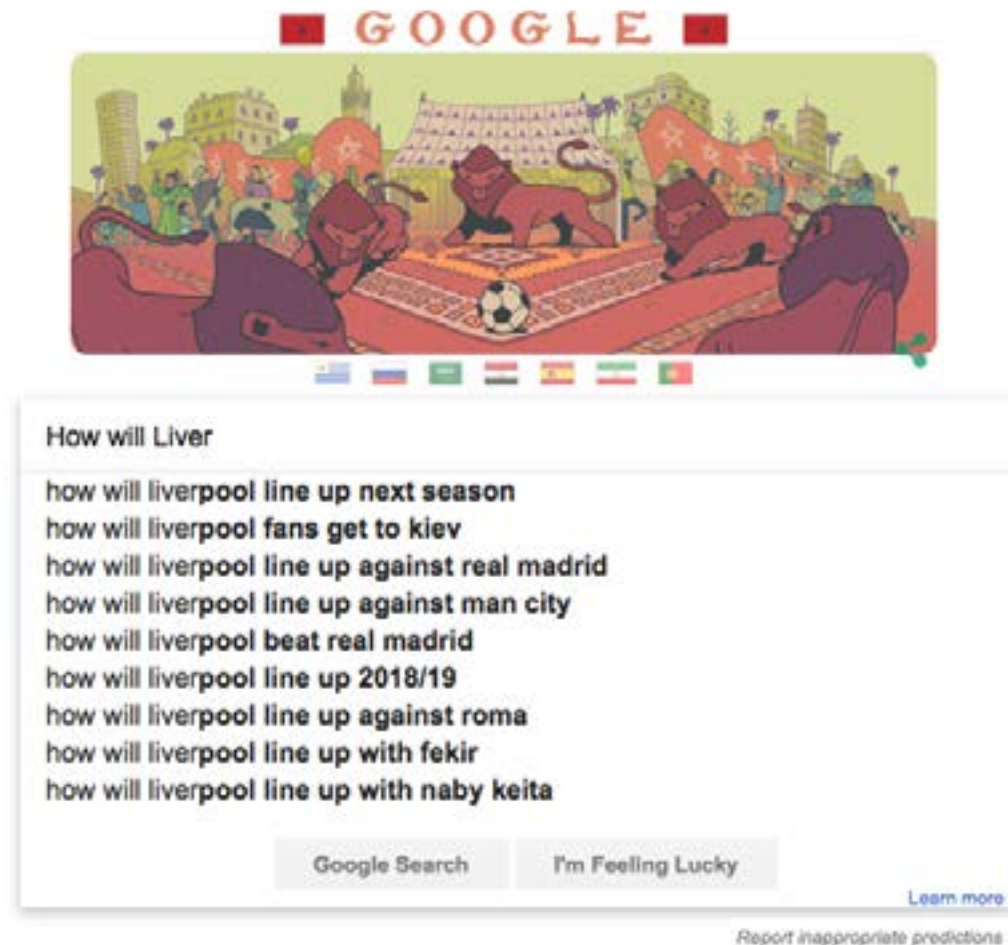
- England
 - America
 - Football
 - Baseball
- 

England	Football	America	Show	Top N: 2
Baseball America				

BMW	car	Boeing	Show	Top N: 2
airplane aircraft				

Word embeddings are everywhere

Lots of uses of context words



Text Embeddings

They can do some 'crazy' things

facebook research

Search



Research Areas

Publications

People

Academic Programs

Downloads & Projects

Careers

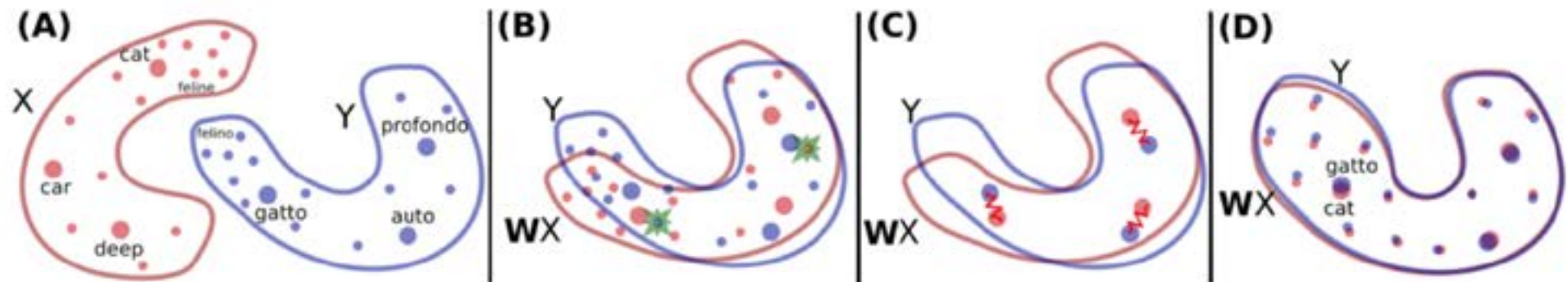
Blog

April 30, 2018

Word Translation Without Parallel Data

International Conference on Learning Representations (ICLR)

By: Alexis Conneau, Guillaume Lample, Marc'Aurelio Ranzato, Ludovic Denoyer, Hervé Jégou



UNCONSCIOUS BIAS IN REVIEWS?

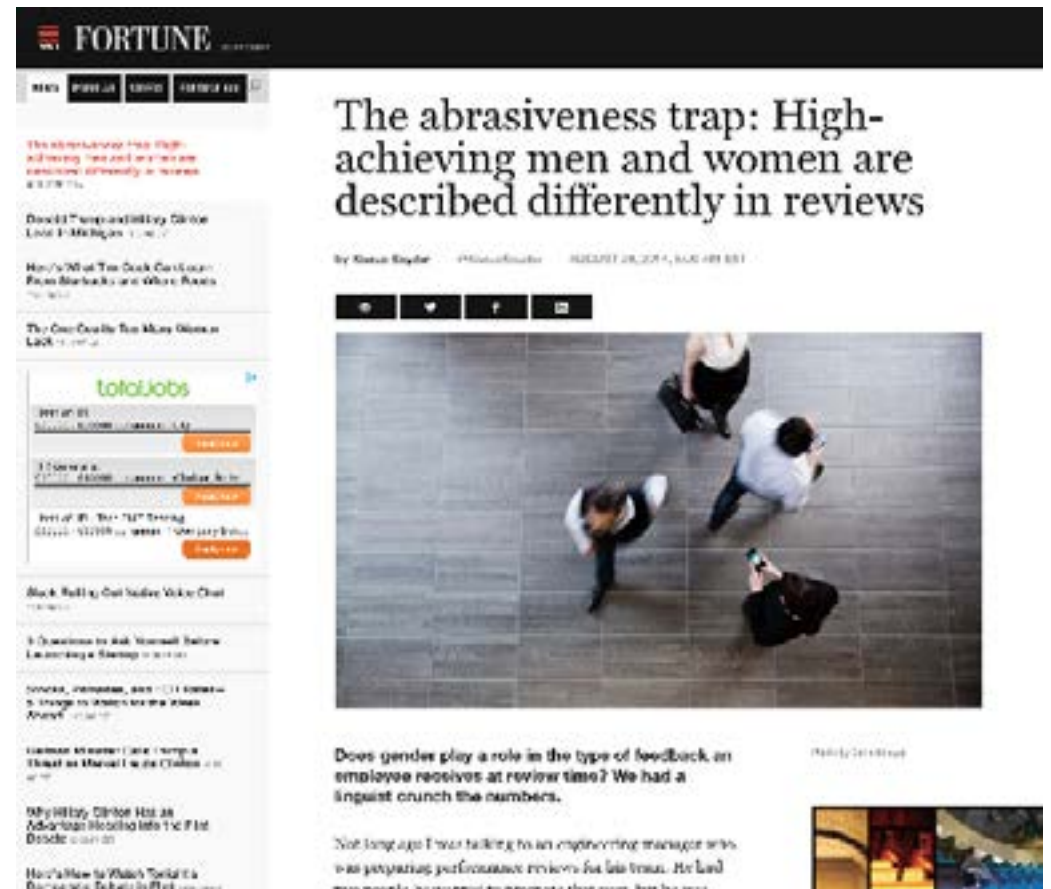
Analysis of the open comments of English
language performance reviews

Introduction

Do managers describe men and women differently in performance reviews?

- Article in Fortune claimed that men and women were described differently in performance reviews
- The sample size was quite small (248 reviews from 108 people)
- Women were given more critical comments
- Words such as 'bossy, abrasive, strident, and aggressive' used for women managers
- Words such as 'emotional and irrational' used when they object.

Are these findings general & do they apply to employee reviews?



Our approach

From the English language texts 10,000 entities were extracted

1

Extensive list of
entities extracted:

Key words

Common phrases

2

Terms ‘controlled’

function

grade

3

Probability
assigned

“What is the
probability that this
entity is more used
about / by women

Conclusion

We couldn't replicate the findings of the Fortune article

There are differences between how managers describe men / women

There are differences between how men / women describe themselves

Research on female & male differences

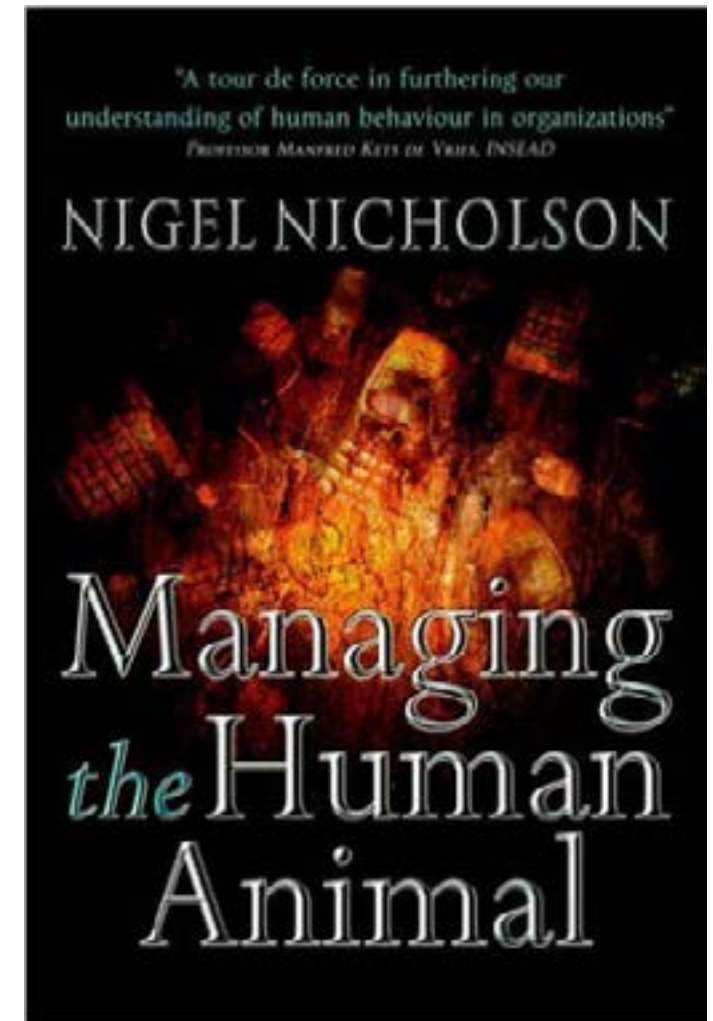
Many of our findings would support existing research

Men...

- are bigger and stronger than women
- don't live as long as women, and lose their mental faculties earlier
- are more competitive than women
- are more self-interested than women
- like to play games more than women
- focus on tasks in a more single-minded fashion than women
- have better mental mapping

Women...

- feel and express emotions more easily than men
- are more empathetic than men
- are more dexterous than men
- are more nurturing than men
- have better social skills than men
- have better verbal skills than men
- are more cooperative than men
- are better at multitasking than men
- have a better memory for

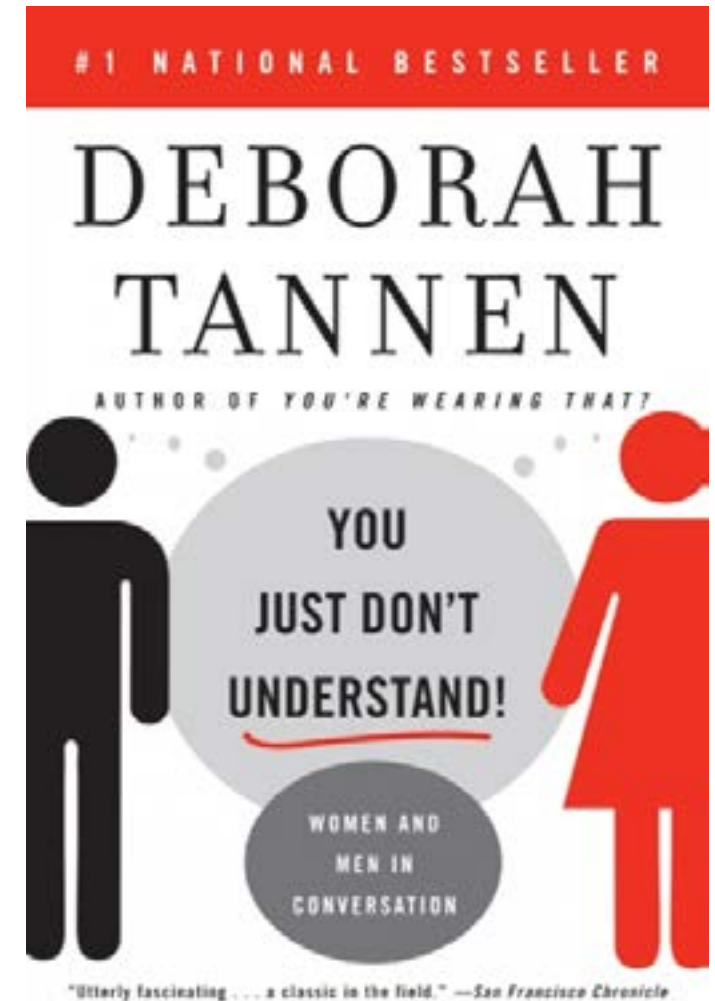


Research on female & male language-use differences

Many of our findings would support existing research

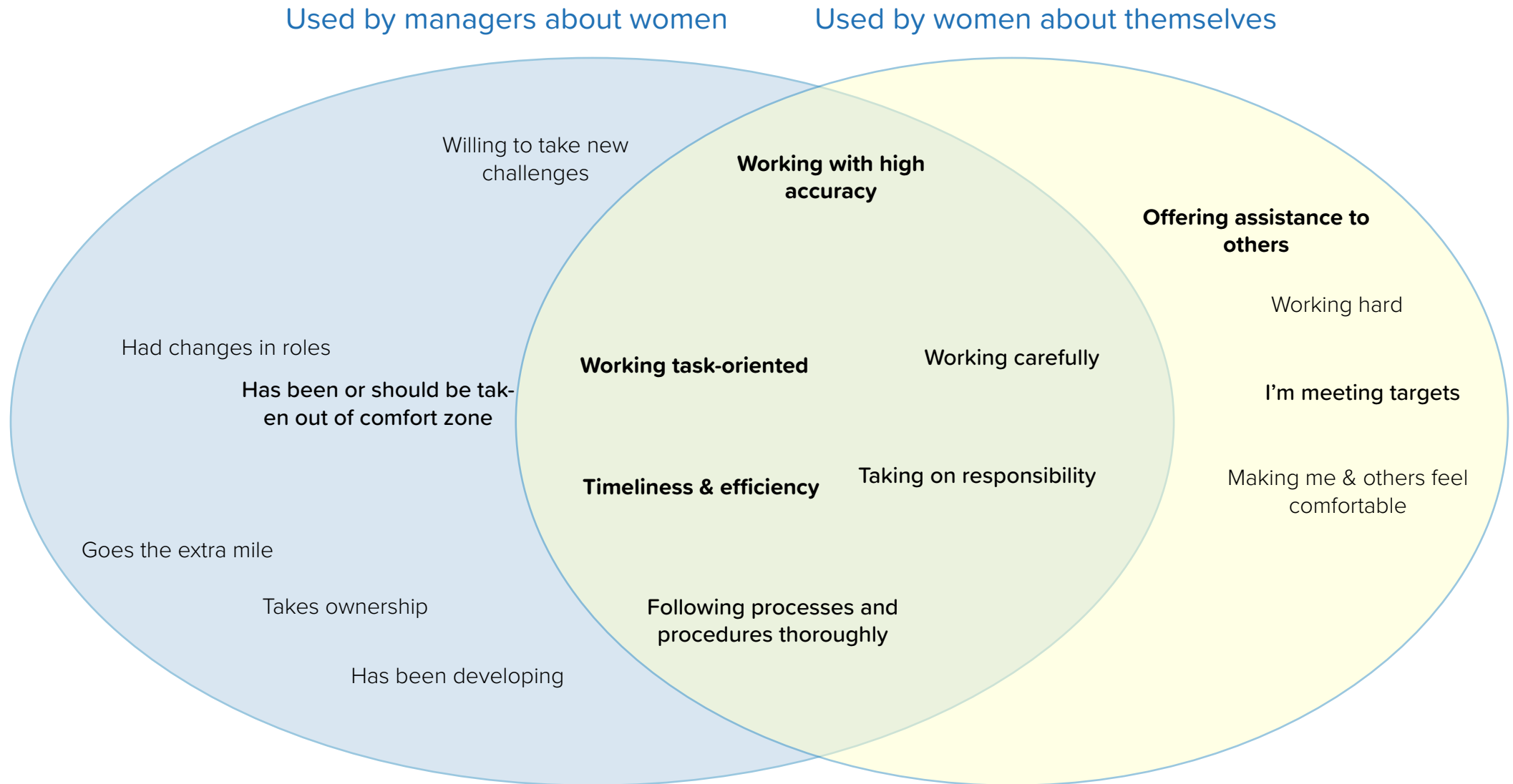
Deborah Tannen's Difference theory

- Status vs Support
- Advice vs Understanding
- Information vs Feelings
- Orders vs Proposals
- Conflict vs Compromise
- Independence vs Intimacy



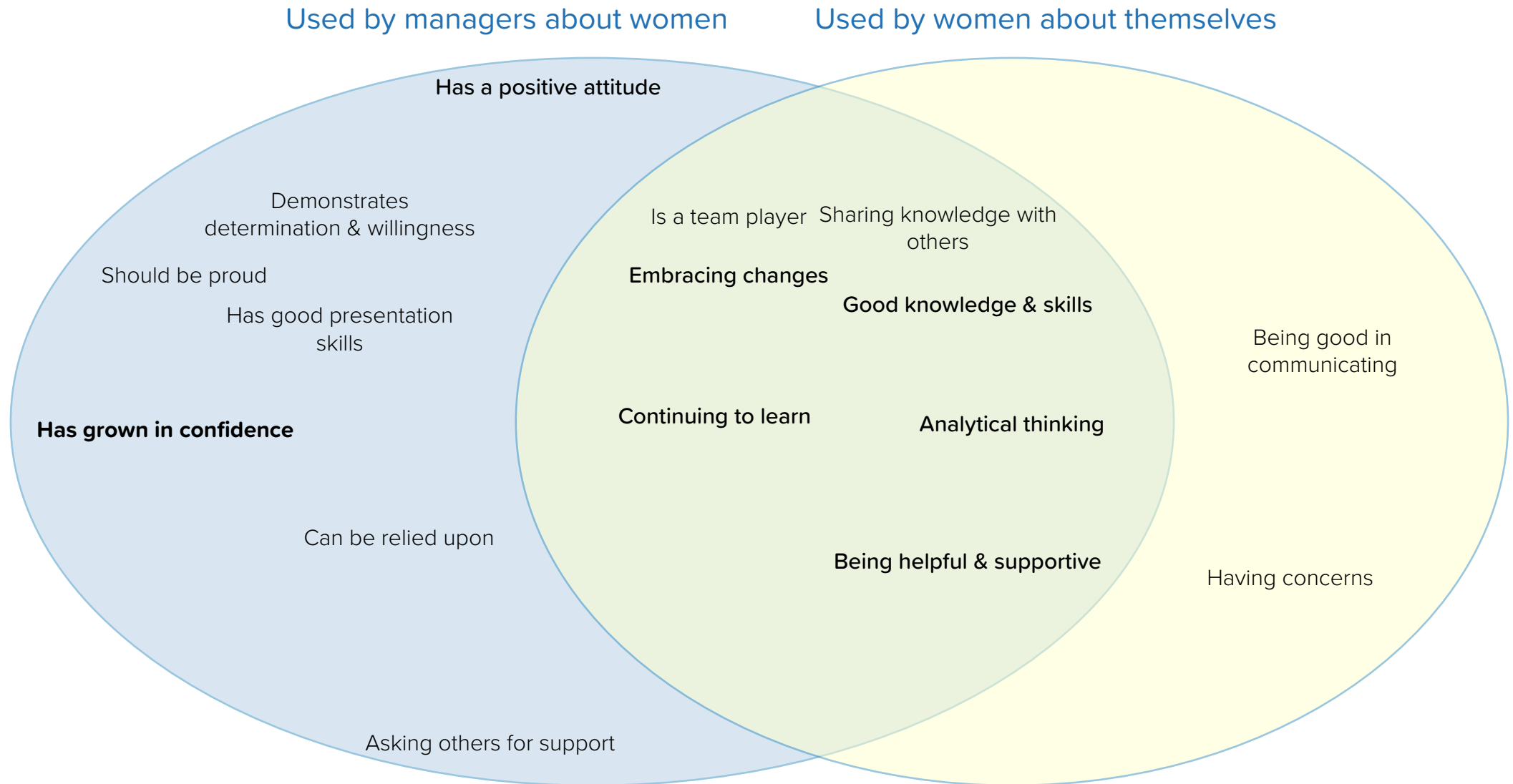
Key themes: what they do - women

Clear differences for several key themes used by managers & women



Key themes: how they do it - women

Clear differences for several key themes used by managers & women

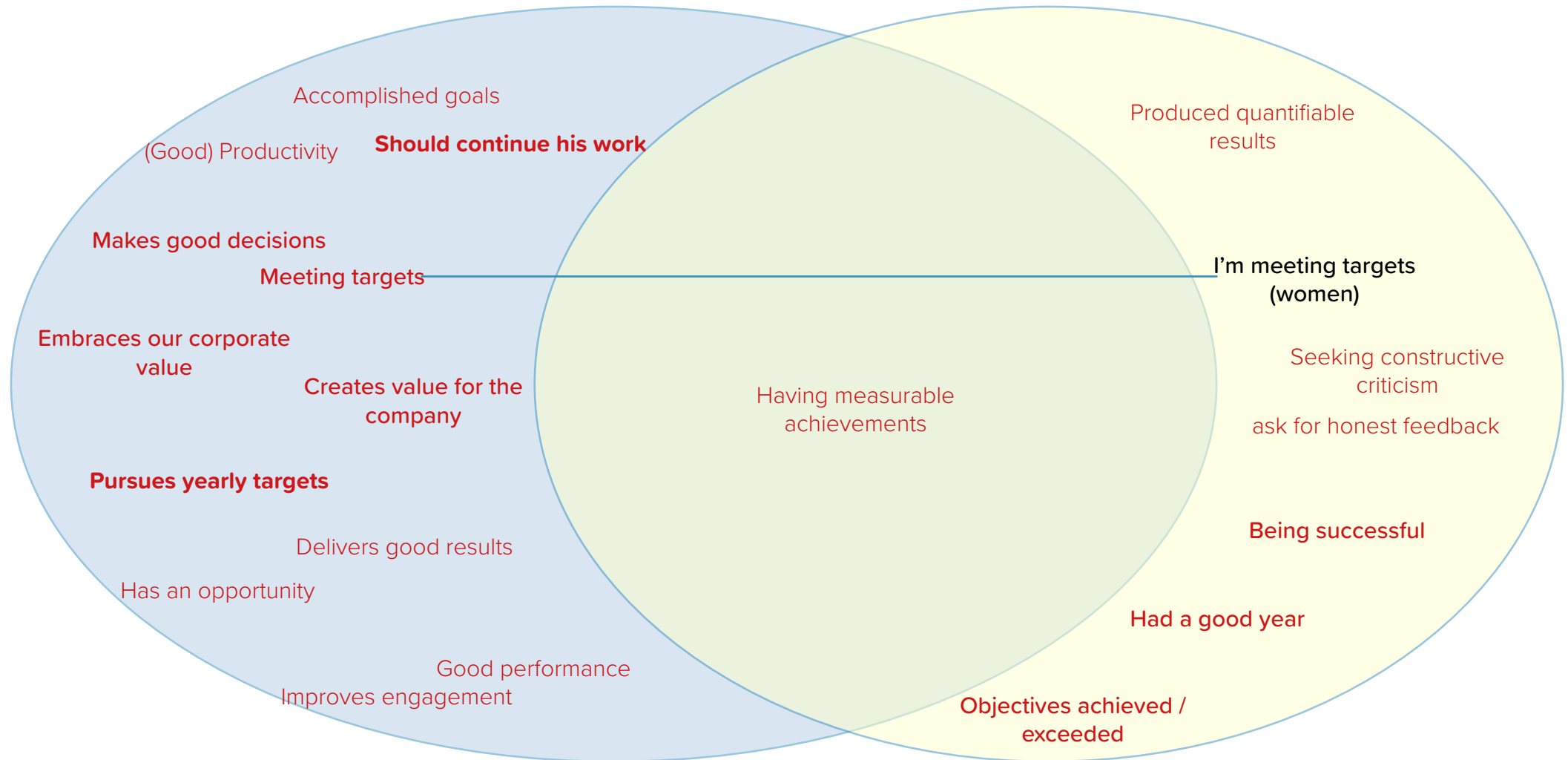


Key themes: what they do - men

A selection of themes more likely to be used by men (black for terms more likely to be used by women)

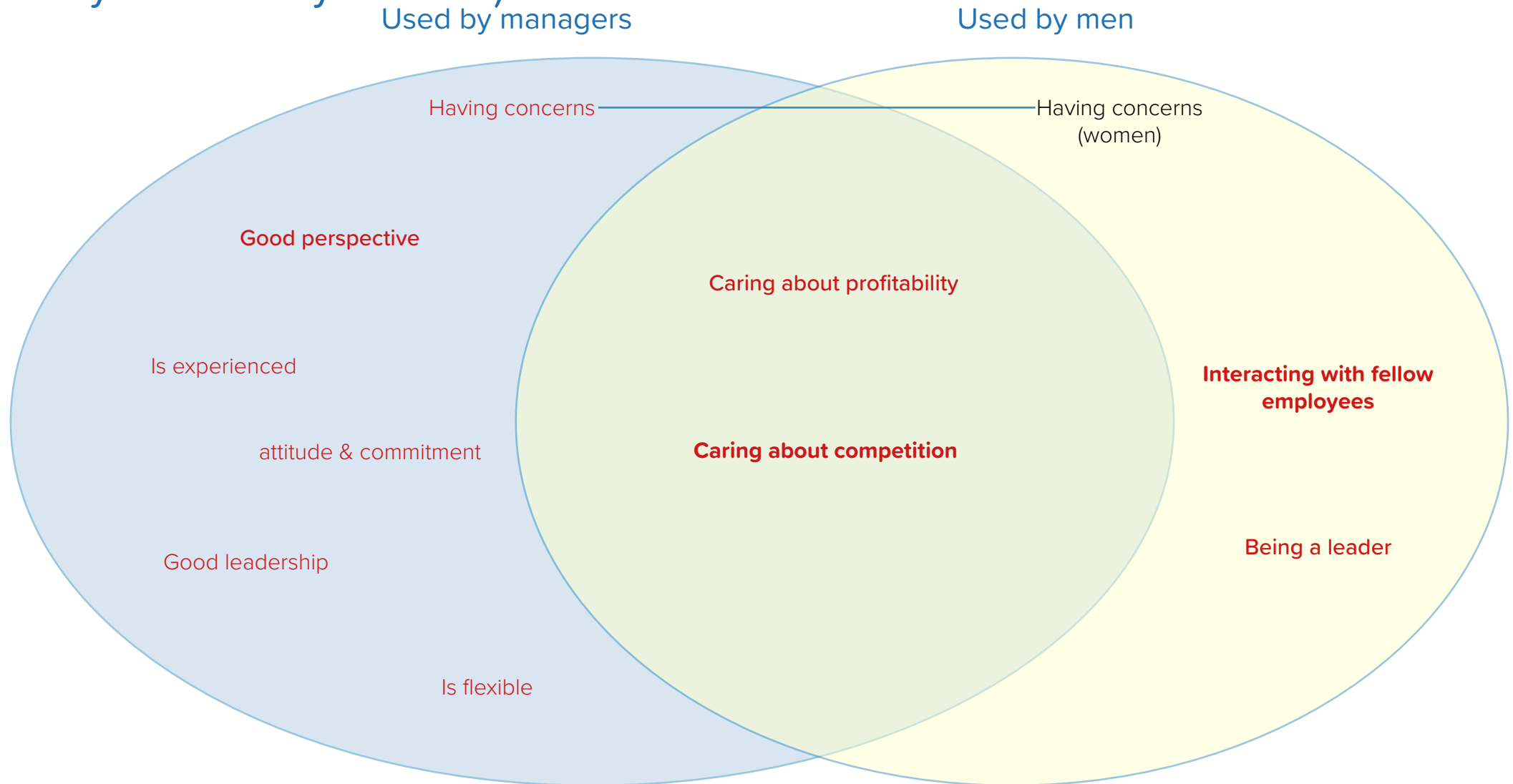
Used by managers about men

Used by men about themselves



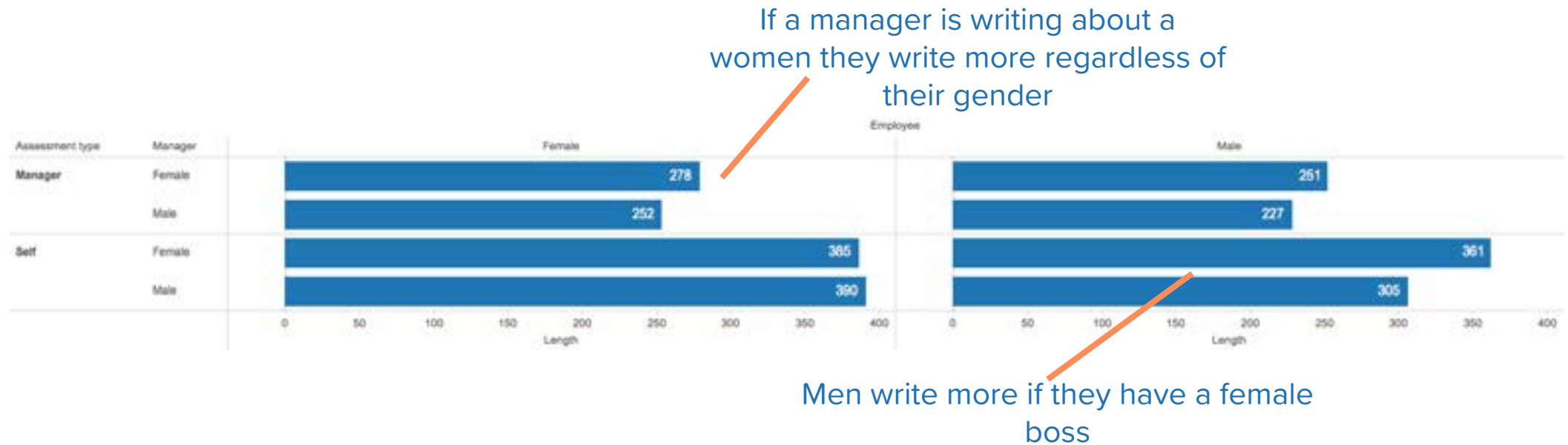
Key themes: how they do it - men

A selection of themes more likely to be used by men (black for terms more likely to be used by women)



If a women is involved, the statements are longer

Regardless of whether it is manager or self-assessment, if a women is involved the text statements are longer.



CONCLUSIONS

CONCLUSIONS

How to get started

Text is very hard to get right.

- Buy vs Build
- If Build, work with domain experts
- Solve very specific problems - the more specific the better
- Most of your time will be cleaning data
- You'll also consume massive resources building training sets
- Don't forget that UI and question makes a **BIG** difference to data quality
- 70% is good, 80% is close to human levels

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